

thermo-photovoltaic cells in solar power systems.

A dual-purpose cloak that scatters electricity and magnetism could also make low-frequency signals clearer. A powered device that accomplishes the same thing could make the technique useful at a more macro level as well. The goal is to find an efficient way to solve the fundamental problems on the undesired electrical-thermal entanglement, for example, turning dissipated heat into a useful source of energy.

This is very exciting work that expands the concept of cloaking to the domain of electrons and, thus, uncovers an interesting approach that may be very useful to thermo-electric applications. Researchers are applying technology developed for the visual cloaking of objects to enable particles hide from passing electronics, which could lead to more efficient thermoelectric devices and new kinds of electronics, thus, making invisibility a key for better electronics.

The new concept could improve the flow of electrons by orders of magnitude and eventually lead to more efficient filters, sensors and thermoelectric devices, and new kinds of electronics. As the components on semiconductors get smaller, the new concept could be used as a better strategy for electron transport. The concept could also lead to a new kind of switches for electronic devices.

Technological advances

Normally, electrons travel through a material in a way that is similar to the motion of electromagnetic waves. In the new electron-cloaking material that has been developed, the process is slightly different. Researchers wanted to carry on further research on how to make some real devices out of this strategy. They developed the idea of harnessing the cloaking mechanisms developed to shield objects from view, applying it

to the movement of electrons.

Researchers are now applying the technology developed for visual cloaking of objects to enable particles to hide from passing electrons. The probability flux of electrons is a representation of the paths of electrons as these pass through an invisible nanoparticle. While the paths are bent as these enter the particle, these are subsequently bent back so that these re-emerge from the other side on the same trajectory these started with—just as if the particle was not there.

The electronic switch could operate by toggling between transparent and opaque states of electrons, thus, turning a flow of these on and off. The concept appears to work in computer simulations, according to the researchers, as they move on to building actual devices to see whether these devices perform as expected.

The initial concept was developed using particles embedded in a normal semiconductor substrate. Now, researchers would like to see if the results can be replicated with other materials, such as 2D sheets of graphene, which might offer interesting additional properties.

They modelled nano-particles with a core of one material and a shell of another. But in this case, rather than bending around the object, electrons actually pass through the particles. Their paths are bent first one way, then back again, so these return to the same trajectory these began with.

The researchers' initial impetus was to optimise the materials used in thermoelectric devices, which produce an electrical current from a temperature gradient. Such devices require a combination of characteristics that are hard to obtain—high electrical conductivity (so that generated current can flow freely), but low thermal conductivity (to maintain a temperature gradient). But the two types of

conductivity need to coexist, so only a few materials offer these contradictory characteristics.

Simulations show this electron-cloaking material could meet these requirements unusually well. The simulations uses particles a few nanometres in size, matching the wavelength of flowing electrons and improving the flow of electrons at particular energy levels by orders of magnitude compared to traditional doping strategies. This might lead to more efficient filters or sensors.

Latest development status

A researcher at University of Texas at Austin, USA, has devised an invisibility cloak that could work over a broad range of frequencies, including visible light and microwaves. This is a significant upgrade from current invisibility cloaks that only cloak a very specific frequency, say, a few hertz in the microwave band, and, more importantly, actually make cloaked objects more visible to other frequencies. This cloak could achieve the goal by being active and electrically-powered, rather than dumb and passive like existing invisibility cloaks.

Another proposed design consists of a conventional meta-material base, but with CMOS negative impedance converters (NICs) placed at the corner of each meta-material square. An NIC is an interesting electronic component that adds negative resistance to a circuit, injecting energy rather than consuming it. NICs are not widely used as researchers are not entirely sure how to use these. This seems to propose that, by interspersing NICs (which must be powered) with the meta-material, multiple frequencies can be cloaked. The proposed cloak is invisible over a large range of frequencies, while a standard passive cloak is only invisible for a small range, and more visible than non-cloaked devices in other ranges. **EFY**